

• SECURITY OPERATIONS — RESEARCH DELIVERABLE

Common Network Security Threats

> DoS/DDoS · MITM · IP Spoofing · DNS Poisoning

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AICTE Oasis Infobyte Internship — Security Analyst Track

TASK REF

TCK-04

DATE

August 2026

SEC.00 Introduction

Network security threats are a critical area of interest for many organizations that operate in the modern digital environment. These entities rely on constant connectivity and trusted data exchange to ensure seamless operations. The more advanced the potential threats are, the more automation is applied to address and neutralize them. At the same time, the ability to distinguish among the most common network security issues and their combined impact on the business is vital for any security-focused organization or individual. In this report, denial-of-service (DoS/DDoS), man-in-the-middle (MITM), IP spoofing, and DNS poisoning types of network security threats will be evaluated in terms of their nature, case scenario, effect, and possible solutions.

SEC.01 DoS / DDoS Attacks

// How It Works

A Denial-of-Service (DoS) attack is designed to overwhelm a targeted server, application, or network with high traffic or request volume, resulting in the system's inability to operate normally. Distributed Denial-of-Service (DDoS) is a particular type of DoS attack, which uses multiple hijacked devices, also referred to as botnets, to flood the target with traffic and bandwidth to take it offline. In most cases, DDoS attacks are executed from networks of hijacked devices without the knowledge of their owners.

DDoS attacks generally fall into three technical categories:

- ▶ Volumetric attacks are characterized by an extremely high traffic volume, which is aimed at saturating the target's bandwidth to create a denial of service (e.g., UDP floods, DNS amplification)
- ▶ Protocol attacks exploit the operating system or network equipment vulnerabilities, resulting in the server's resource exhaustion (e.g., SYN floods)
- ▶ Application-layer attacks target by exploiting weaknesses in the target application, which can be challenging to mitigate

// Case File — Mirai Botnet vs. Dyn (21 October 2016)

The Mirai botnet was created by scanning the Internet for IoT devices (routers, IP cameras, and DVRs) with default credentials, which were compromised by malware to form a large-scale botnet of approximately 145,000 devices. On 21 October 2016, the attackers launched an attack against the DNS provider Dyn by overwhelming its servers with tens of millions of malicious companies and services were using the Dyn platform for domain name resolution, Twitter, Netflix, PayPal, Reddit, Amazon, Spotify, and other popular services were temporarily unavailable in the United States and parts of Europe. This outrage occurred at approximately 06:10:00 UTC and was resolved at around 08:10:00 UTC for the first time; however, two smaller attacks occurred throughout the day.

// Impact

\$22K / MIN DOWNTIME **51%** REVENUE LOSS RATE **\$2.5M** AVG RECOVERY COST

- ▶ Dyn lost approximately 14,000 domains after the incident due to the loss of 8% of their user base, which could permanently affect their revenue, as a company's reputation is paramount.

// Mitigations

- ▶ Continuous monitoring of traffic levels against baselines for spikes
- ▶ Use of Content Delivery Networks (CDNs) and cloud-based traffic scrubbing services (e.g., Cloudflare, AWS Shield)
- ▶ Resilience in hosting infrastructure – segmentation of services, provisioned headroom, geographic diversity
- ▶ Collaboration with upstream ISP to perform filtering as close to the source as possible
- ▶ Hardening IoT and device hardening against attack, including changing default credentials to reduce attack surface

SEC.02 Man-in-the-Middle (MITM) Attacks

// How It Works

A MITM attack is when a perpetrator manages to position themselves between two people or parties communicating, and they are able to intercept and possibly alter messages between them, while both parties are led to believe that they are communicating with each other directly. The MITM attacker does this by impersonating each party sufficiently to be able to bypass security measures designed to prevent such attacks.

Common MITM techniques include:

- ▶ ARP spoofing: This involves spoofing address resolution protocol messages and is used mainly in local area networks.
- ▶ DNS spoofing/poisoning: This is when the attacker corrupts the Domain Name System database to redirect victims to fake websites.
- ▶ SSL/TLS stripping: This is the non-secure ones, and it is usually done to circumvent the encryption and authentication mechanisms of HTTPS.
- ▶ Evil twin: This technique uses rouge Wi-Fi access points to lure victim into connecting to them.
- ▶ Session hijacking: This is when an attacker takes over a legitimate user session by stealing their session token or cookie.
- ▶ Fraudulent certificate authorities: This is when attackers use forged CAs to bypass the encryption and authentication mechanisms of HTTPS.

// Case File — DigiNotar CA Breach (2011)

A Dutch certificate authority, DigiNotar was hacked, and false certificates for various companies, including Google, were issued. Since the certificates were trusted by browsers, attackers could impersonate secure connections to Google and other companies and intercept data. In particular, it was reported that many Iranians were victimized through this scheme, which was discovered when browsers started rejecting DigiNotar certificates. This security breach significantly impacted the company, which eventually filed for bankruptcy because it lost its primary authority as a certificate provider after browsers and users stopped trusting it.

// Impact

- ▶ Victims' supposedly encrypted traffic, including Gmail communications, was silently intercepted and readable by attackers
- ▶ Trust in the certificate authority system was shaken industry-wide
- ▶ DigiNotar ceased operating entirely within weeks of disclosure

// Mitigations

- ▶ HTTPS/TLS encryption everywhere, using asymmetric encryption to protect data in transit
- ▶ DNS over HTTPS (DoH) or DNS over TLS (DoT) to encrypt DNS queries themselves
- ▶ HTTP Strict Transport Security (HSTS), paired with the HSTS preload list, to prevent SSL-stripping
- ▶ VPN usage on public or unsecured networks
- ▶ Use WPA3 Wi-Fi encryption to protect devices communicating over wireless access points (AES with up to 256-bit and SHA-384)
- ▶ Strong password policy and multi-factor authentication
- ▶ Employee/user education regarding threats and best practices in cybersecurity
- ▶ Implement zero-trust architectures that require all devices/users to be authenticated and authorized before being granted access to resources, rather than trusting devices on a private network
- ▶ Continuous network monitoring for anomalies such as unexpected delays or disconnections

SEC.03 IP Spoofing

// How It Works

IP spoofing is a security attack where the source address in an internet packet is altered in order to conceal the origin of the traffic initiator. Since IP addresses provide a computer's identity on the internet, successful spoofing allows an entity to impersonate another computer and, therefore, bypass IP-based security measures or hide their actions behind another system's address. In practice, IP spoofing is rarely used as a standalone attack. Instead, it is used in synergy with other threats:

- ▶ Volumetric DDoS attacks: spoofed source IPs allow attackers to obscure a botnet's true origin and amplify the attack's power through traffic reflection.

- ▶ MITM attackers can use IP spoofing to impersonate a compromised server and interject themselves in the middle of a communication session.
- ▶ TCP handshake hijacking is when an attacker can forge the destination IP address during the TCP 3-way handshake and spoofed SYN / SYN-ACK / ACK message and receive response traffic intended for a legitimate server.

// Case File — Florida Banks DNS/IP Spoofing (2006)

Attackers compromised ISP gateways of three Florida-based banks and redirected their customer traffic to a domain controlled by malicious actors. As a result, customers were tricked into thinking they were interacting with their bank when, in reality, all their credentials, including ATM PINs, CVV codes, and card numbers, were being siphoned off to the attacker's web server. DNS spoofing in this instance allowed attackers to facilitate large-scale financial fraud by tricking victims into providing sensitive data.

// Impact

- ▶ Direct fraud: Theft of financial credentials from customers of the targeted banks.
- ▶ Damage to reputation: Customers may become reluctant to use the bank's online services in the future.
- ▶ Disclosure of PII: Spoofing DNS responses allowed attackers to intercept and exfiltrate sensitive data from users who believed they were typing it into the bank's official website.

// Mitigations

- ▶ Ingress and egress packet filtering at the firewall level based on access control lists can prevent packets with spoofed IP addresses from entering or exiting a network.
- ▶ Behavioral packet filtering similar to the above, but applied in real-time by a firewall or intrusion prevention system based on irregular patterns observed in the traffic.
- ▶ Secure communications transport layer encryption protocols (HTTPS/TLS) ensure confidentiality of communications, even if packets are spoofed.

SEC.04 DNS Poisoning / Spoofing — Bonus

// How It Works

DNS poisoning attacks DNS servers or resolvers, altering the records those services store so that a domain name points to an incorrect IP address. By interfering with the critical infrastructure that makes up the internet's directory, attackers can lure victims to malicious websites by tricking them into visiting seemingly legitimate URLs that appear in the address bar.

// Case File — Malaysia Airlines DNS Spoofing (2015)

The attackers spoofed the DNS records of Malaysia Airlines' official website, causing any request to the domain to return an incorrect, attacker-controlled IP address. The attack did not directly steal any customers to book or change flights.

// Impact

- ▶ The ability to conduct business may be interrupted if the spoofed records point to an illegitimate or non-existent server.
- ▶ Users who unknowingly visit the spoofed domain may feel tricked or misled, damaging the attacked entity's reputation.
- ▶ Similar attacks that use spoofed login pages to harvest customer credentials could be considerably more damaging, as in the Florida bank's DNS poisoning case.

// Mitigations

- ▶ Using DNSSEC will allow DNS resolvers to check if the received records were signed with an authentic public key.
- ▶ Encrypting DNS queries using HTTPS (DoH) or TLS (DoT) protocols prevents attackers from reading or altering them.
- ▶ Clearing DNS caches and regularly checking for suspicious records can limit the damage of most poisoning attacks.

SEC.05 Comparison Matrix

THREAT	ATTACK VECTOR	WHO'S AT RISK	DIFFICULTY	MITIGATION EASE
DoS / DDoS	Traffic/resource flooding via botnets	Any internet-facing service; e-commerce, DNS providers	LOW-MED	MEDIUM
MITM	Interception between two communicating parties	Users on public/unsecured Wi-Fi; unencrypted traffic	MEDIUM	MED-HIGH
IP Spoofing	Forged packet source addresses	Weakly-filtered networks; enabler for DDoS/MITM	MEDIUM	MEDIUM
DNS Poisoning	Corrupting DNS resolution records	Any user relying on standard DNS	MED-HIGH	HIGH

SEC.06 Conclusion

Three key takeaways for a network administrator:

- ▶ **Availability and confidentiality are both attack surfaces.** The Dos/DDoS attacks take away the system's availability, while MITM, IP spoofing, and DNS poisoning compromise confidentiality by intercepting or altering data. A complete security solution should address both aspects of network security.
- ▶ **Encryption and verification are the common underlying principles,** which are used in HTTPS/TLS, DNSSEC, DoH/DoT, packet filtering, and other security measures. All these technologies address the same issue, securing communication by ensuring the trustworthiness of the other party.

- ▶ **These threats compound rather than occur in isolation.** IP spoofing can be used to launch DDoS attacks and MITM attacks, whereas DNS poisoning can lead to IP spoofing and other types of deception. Thus, defending against one type of attack might be insufficient, as several types can be combined.

SEC.07 References

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